

STATEMENT: Implement a real-time event processing system using a Queue data structure. The system

Add an Event: When a new event occurs, it should be added to the event queue.

- Process the Next Event: The system should process and remove the event that has been in the queue the longest.
- Display Pending Events: Show all the events currently waiting to be processed.
- Cancel an Event: An event can be canceled if it has not been processed.

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```
from collections import deque;

class Q_deque:

    def __init__(self):
        self.q = deque()

    def add_event(self,event):
        self.q.append(event)
        print(f"Event '{event}' added")

    def process_event(self):
        if self.q:
            event = self.q.popleft()
            print(f"Event is processed {event}")
        else :
            print(f"Event {event} is already processed ")

    def cancel_event(self,event):
        if event in self.q:
            self.q.remove(event)
```

```
print(f"Event '{event}' is remove")
print(self.q)
else:
print(f"Event '{event}' is canceled")
def display(self):
print(f"Event is display'{self.q}'")
def menu(self):
while True:
print("===menu===")
print("1. added event:")
print("2. process event:")
print("3. cancel event:")
print("4. display event:")
print("5. Exist:")
choice = int(input("Enter a number:"))
if (choice == 1):
event = input("Enter a event:")
self.add_event(event)
elif (choice == 2):
self.process_event()
elif (choice == 3):
event=input("enter event to be delete:-")
self.cancel_event(event)
elif (choice == 4):
self.display()
elif (choice ==5):
```

```
print("Exit")  
break;  
obj = Q_deque()  
obj.menu()
```

OUTPUT:

===menu===

1. added event:
2. process event:
3. cancel event:
4. display event:
5. Exist:

Enter a number:1

Enter a event:E1

Event 'E1' added

===menu===

1. added event:
2. process event:
3. cancel event:
4. display event:
5. Exist:

Enter a number:1

Enter a event:E2

Event 'E2' added

===menu===

1. added event:

2. process event:

3. cancel event:

4. display event:

5. Exist:

Enter a number:1

Enter a event:E3

Event 'E3' added

===menu===

1. added event:

2. process event:

3. cancel event:

4. display event:

5. Exist:

Enter a number:2

Event is processed E1

===menu===

1. added event:

2. process event:

3. cancel event:

4. display event:

5. Exist:

Enter a number:4

Event is display'deque(['E2', 'E3'])'

===menu===

1. added event:

2. process event:

3. cancel event:

4. display event:

5. Exist:

Enter a number:3

enter event to be delete:-E1

Event 'E1' is canceled

===menu===

1. added event:

2. process event:

3. cancel event:

4. display event:

5. Exist:

Enter a number:5

Exit

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